

Algorithms and Certificates for Finite Effective Causal Descent: Width-Based Synthesis for Sensor-Limited Mechanical Control

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Abstract

Finite decision systems under partial observation face a simple obstruction: a controller cannot choose different actions in scenarios that its sensors do not distinguish. This article develops the finite algorithmic companion to effective causal descent: a proof-producing synthesis pipeline for finite sensor-limited mechanical models. A finite presentation consists of policy variables, finite domains, and explicit local relations; global policies are exactly the assignments satisfying all relations. Effectivity and bounded realisability are represented by finite implementation catalogues, costs, and budgets. The main algorithmic layer is a certificate-producing dynamic program over tree decompositions, with ordinary and weighted-width bounds, independently checkable message tables, and typed certificates for policies, refutations, optimum costs, fibre conflicts, sensor repairs, finite-element rows, and sparsification ladders. The mechanics layer converts declared structural models with rational data into certified exact influence rows, and turns certified affine response rows into inner and outer finite ECD models, proving the sound sandwich $\Gamma(P^-) \subseteq \Gamma(P^{\text{true}}) \subseteq \Gamma(P^+)$. Width-adaptive sparsification gives monotone certified refinement rather than a performance claim. A recorded computational study executes a mechanics benchmark suite over truss and mass-spring families with quantised-sensor observation variants, a width-adaptive sparsification study over retained-scope ladders, and a comparative scaling study of the certified pipeline against external SAT, CP-SAT, and MILP solvers on identical mechanics-derived instances; a version-exact replication reproduces the recorded comparative study, and a dense-family extension to seventeen and eighteen actuators records the certificate-growth and memory boundary of the pipeline; all limits, relation-table boundaries, and unresolved brackets are reported explicitly. An exact two-spring demonstrator and a finite prototype artifact exercise the certificate grammar, reject corrupted certificates, and produce controlled-width benchmark reports. In every external comparison, an answer is accepted only when cross-checked by an independently verified native certificate, and timeouts, missing solvers, and solver statuses without a certified claim remain inconclusive.

Keywords: finite effective causal descent; sensor-limited control; certifying algorithms; tree decompositions; weighted width; finite-element certification; sparsification; independent verification.

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1 Introduction

A mechanically admissible response need not be implementable by a controller whose observations fail to distinguish the situations requiring different actions. A load case, damage state, or inspection record may determine which actions are safe, while the available sensor readings merge several such cases into one information state. Once the information state is fixed, the controller has only one command available. The first task is therefore not to solve a continuous mechanics problem in full generality, but to certify the finite relation between observations, admissible actions, and implementable policies in a declared model.

This article develops the finite, proof-producing side of effective causal descent (ECD). The foundational monograph (version 47) supplies the presentation-relative language: local validity,

compatibility of a diagram, effective realisability, and bounded realisability. Here we specialise that language to explicit finite instances and to mechanically generated finite relations. The guiding rule is: *topological order is the constraint; feasibility is the priority*. We first fix the finite input representation, then the certificate grammar, then the one-shot fibre and sensor-repair theory, then the solver and verifier, then the mechanical relation generator, then the sparsification and benchmark layers. Comparative performance claims come last and require experiments; they are not consequences of the structural theory.

The contribution is organised around certificates. A successful run should return a policy that an independent verifier checks. A negative run should return a refutation certificate, not merely a solver log. A mechanical row should carry the numerical or exact-arithmetic evidence justifying its safe, unsafe, or unresolved status. A sparsified instance should come with interval bounds proving the appropriate inner–outer inclusions. The computational statuses are deliberately limited: certified satisfiable, certified unsatisfiable, certified optimal, or inconclusive. The design follows the certifying-algorithms discipline: every output is checkable evidence, not a trusted narrative [19].

The finite contributions are the following. First, we fix an explicit input schema for finite ECD presentations and implementation catalogues. Second, we specify typed certificates that an independent verifier can check. Third, we prove one-shot fibre conflict bounds, reachability-correct causal compilation, and explicit-table complexity boundaries. Fourth, we give a width-based dynamic program with weighted-width accounting and message-table certificates. Fifth, we characterise one-shot sensor repair exactly as a hitting-set condition on inclusion-minimal conflict cores, and prove the design side hard: minimum-cardinality repair from a listed candidate family is NP-complete and W[2]-hard in the number of added sensors; the unrestricted formulation is classified alongside — degenerate under free sensor alphabets, and under a fixed value alphabet an exact reduction to per-fibre transversal bounds with the same hardnesses. Sixth, we connect exact affine mechanical rows to finite inner and outer ECD models. Seventh, we prove a monotone sparsification ladder for retained scopes. Eighth, we convert declared structural models with rational data into certified exact influence rows whose base states are verified by exact residual substitution. Finally, we report an artifact that exercises these certificates, rejects corrupted evidence, and produces controlled-width benchmark summaries; the recorded mechanics study executes a benchmark suite with quantised-sensor observation variants, a sparsification ladder study measuring the width–inconclusiveness tradeoff, and a comparative scaling experiment against external solvers on identical instances, without claiming external-solver superiority.

Relation to prior work

The machinery is assembled from established lines. Acyclic join processing supplies the width-zero regime [26]; width-directed consistency and decomposition supply the bounded-width regime [11, 8, 5]; the certifying-algorithms discipline requires outputs that carry checkable evidence [19]; parameterised complexity calibrates what width-bounded computation cannot achieve [10, 14]. The source problem is posed by the decentralised-control tradition: synthesis and decision making under nonclassical information structures [25, 21, 3]. Sensor-selection studies budget observations for information value [17], and nested-dissection eliminations price fill-reducing orders on meshes [13, 18]. None of these lines supplies the assembly this article reports: typed certificates for finite ECD instances — negative answers as verified refutations rather than solver logs, sensor repair as a hitting-set certificate, mechanical rows as exact-arithmetic evidence, and sparsification as verified inner–outer inclusions — under a status discipline that records inconclusive outcomes instead of discarding them. The engines are classical; the assembly and its recorded execution are the contribution.

2 Finite ECD presentations

Definition 2.1 (Finite ECD presentation). A finite ECD presentation is a tuple

$$\mathcal{P} = (X, (D_x)_{x \in X}, (S_i, R_i)_{i \in M}),$$

where X is a finite set of policy variables, each D_x is a finite nonempty domain, each scope $S_i \subseteq X$ is finite, and

$$R_i \subseteq \prod_{x \in S_i} D_x$$

is an explicitly represented permitted table. Its global policy space is

$$\Gamma(\mathcal{P}) = \{a \in \prod_{x \in X} D_x : a|_{S_i} \in R_i \text{ for all } i \in M\}.$$

In a sensor-limited mechanics model, a variable often has the form (j, o) , where j is an actuator or decision channel and o is an information state produced by the available sensors. The finite relation tables record which joint commands are certified safe for each scenario or local mechanical constraint.

Definition 2.2 (Canonical finite input). The verifier-facing input convention is

$$I = (X, (D_x)_{x \in X}, (S_i, R_i)_{i \in M}, O, \text{obs}, \mathcal{L}, c, B).$$

Here X , D_x , S_i , and R_i give the explicit finite relation instance; O records observation states and $\text{obs} : W \rightarrow O$ records the map from a finite scenario set W , or from histories, to information states when those data are relevant; $\mathcal{L}$ is an optional finite implementation catalogue; c is its cost map; and B is an optional budget. If a table is replaced by a succinct circuit, automaton, or finite-element generator, the input representation has changed and the corresponding complexity or soundness claim must be restated for that source representation.

Definition 2.3 (Finite implementation catalogue). For controller j , let $\mathcal{L}_j$ be a finite catalogue of implementations with evaluation map $\text{eval}_j(\ell, o)$, defined on the controller's information states $o \in O_j$, and cost $c_j(\ell)$. Introduce implementation variables $\ell_j \in \mathcal{L}_j \cup \{\perp\}$, where $\perp$ is compatible with every behavioural assignment and has infinite cost. A finite-cost catalogue policy is one satisfying

$$x_{j,o} = \text{eval}_j(\ell_j, o) \quad (o \in O_j)$$

with every $\ell_j \neq \perp$.

Thus finite set-theoretic existence is separated from existence in a declared implementation catalogue, and budgeted existence is separated again by a cost bound. This is the finite analogue of the nested spaces $\Gamma_{\text{set}} \supseteq \Gamma_{\text{eff}} \supseteq \Gamma_{\mathcal{B}}$, but it is entirely explicit.

Proposition 2.4 (Finite effective collapse). *Let $\mathcal{P}$ be a finite ECD presentation whose variables carry the form $x_{j,o}$, where j ranges over a finite set of controllers with information states $o \in O_j$, the variables of each controller sharing a domain D_j . Equip each controller with the unrestricted catalogue $\mathcal{L}_j$ of all total functions $O_j \rightarrow D_j$, evaluated by $\text{eval}_j(\ell, o) = \ell(o)$. Then every policy $a \in \Gamma(\mathcal{P})$ is a catalogue policy, so set-theoretic and unrestricted-catalogue existence coincide.*

Proof. Every catalogue policy is in particular an assignment satisfying every row, so catalogue existence implies set-theoretic existence. Conversely, given $a \in \Gamma(\mathcal{P})$, put $\ell_j(o) = a_{j,o}$ for $o \in O_j$; then $\ell_j \in \mathcal{L}_j$ and $a_{j,o} = \text{eval}_j(\ell_j, o)$ for every o , so a is a catalogue policy with every $\ell_j \neq \perp$. $\square$

Remark 2.5 (What the collapse separates). The collapse confines the finite analogue of $\Gamma_{\text{set}} \supseteq \Gamma_{\text{eff}} \supseteq \Gamma_{\mathcal{B}}$: under unrestricted catalogues the set-theoretic and effective questions coincide, and the remaining separations are the declared-catalogue and budget separations of Definition 2.2. A bounded class retains content only after the bound is declared — controller memory, representation size, or cost budget. Existence is thereby separated from the cost of finding it: a synthesis timeout or a table-size limit is inconclusive, not evidence of non-existence, which is the status discipline of Section 3.

Remark 2.6 (Value costs and catalogue costs). Two cost notions coexist and must not be conflated. The catalogue cost map c of Definition 2.2 prices implementations $\ell \in \mathcal{L}_j$. Optimum certificates instead use value costs: a costed instance additionally carries nonnegative rational value costs $\kappa_x(v)$ for $x \in X$ and $v \in D_x$, and the cost of an assignment is

$$\kappa(a) = \sum_{x \in X} \kappa_x(a_x).$$

The DP-OPTIMUM certificates introduced in Section 3 and developed in Section 6 certify the minimum of κ over $\Gamma(\mathcal{P})$; the benchmark suite of Section 11 uses per-actuator value costs with $\kappa_x(1) = 1$ and $\kappa_x(0) = 0$, so its optima report the minimum number of active actuators.

3 Certificate grammar and status discipline

The verifier accepts typed certificates rather than solver narratives. The grammar contains the following core forms.

Certificate	Data	Certified claim
POLICY	total assignment to all variables	global section
FIBRE-CONFLICT	one observation fibre and excluded-action witnesses	no one-shot selector on that fibre
SENSOR-REPAIR	selected sensors and conflict cores	sensor sufficiency or optimality in a finite master problem
TREE-DECOMP	bags, edges, root, factor placement	valid decomposition
DP-REFUTATION	decomposition and message tables	no global assignment
DP-OPTIMUM	decomposition, message tables, root value	minimum finite cost
FE-ROW	response interval and row status	safe, unsafe, or unresolved finite row
SPARSIFICATION-BATCH	retained scopes and omitted-term bounds	inner and outer sparse model

Remark 3.1 (Status discipline). The implementation reports only four mathematical statuses: CERTIFIED-SAT, CERTIFIED-UNSAT, CERTIFIED-OPT, and INCONCLUSIVE. A timeout, missing external solver, failed certificate verification, or unresolved numerical row is inconclusive. It is never a proof of infeasibility. The verification gate additionally emits bookkeeping markers for the state of a gate check — VERIFIED, REJECTED, COMPLETE, and RELATION-TABLE-LIMIT for rows whose relation enumeration exceeds the recorded cap; these are harness statuses, not answers, and the four mathematical statuses above are unaffected by them.

4 One-shot fibres and causal compilation

The FIBRE-CONFLICT certificate type of the grammar rests on the observation fibres. Writing $o = \text{obs}$ for the observation map, the observation fibre over an observation value $z \in O$ is $o^{-1}(z) = \{w \in W : o(w) = z\}$: the set of scenarios that the available observation map does not distinguish from one another.

Theorem 4.1 (One-shot fibre criterion). *Let $o : W \rightarrow O$ be an observation map, let A be a finite action set, and let $S_w \subseteq A$ be the actions admissible in scenario w . An observation-uniform action selector exists if and only if*

$$A_z := \bigcap_{w:o(w)=z} S_w \neq \emptyset$$

for every realised observation value z . If a fibre fails, a certificate using at most $|A|$ scenarios in that fibre excludes every action, and the bound is tight.

Proof. A selector chooses one action a_z for every observation value. The action is valid on the fibre over z exactly when $a_z \in S_w$ for every scenario in that fibre, equivalently $a_z \in A_z$. If $A_z = \emptyset$, choose for each action $a \in A$ one scenario w_a in the fibre with $a \notin S_{w_a}$. The distinct chosen scenarios exclude all actions and have cardinality at most $|A|$. Tightness is witnessed by $S_{w_j} = A \setminus \{j\}$ for $A = \{1, \dots, k\}$. $\square$

Theorem 4.2 (Reachability-correct forbidden-tuple compilation). *Consider a finite explicit history tree whose predictor nodes are partitioned into information cells. Introduce one policy variable for each information cell. For each unsafe leaf, read the information-cell/action labels along the root-to-leaf path. If the path requires the same information cell to take two different actions, discard it. Otherwise add the corresponding forbidden tuple. The resulting finite presentation has global assignments exactly the policies that avoid every unsafe leaf.*

Proof. A path requiring two different actions at one information cell is unrealisable by any observation-uniform policy, so it contributes no feasible unsafe execution. Every remaining unsafe path is reached precisely when the policy matches all action labels on that path; forbidding the resulting tuple is therefore exactly the condition that this unsafe leaf is not reached. $\square$

5 Finite complexity boundary

The explicit-table representation fixes the complexity claims. The basic frontier is standard finite CSP complexity, but it is important to state it in the same encoding used by the verifier.

Problem	Complexity
Verify a proposed assignment	P
Unary constraints	P
Boolean binary constraints	P via 2-SAT
Boolean ternary explicit-table constraints	NP-complete
Binary disequality over a three-element domain	NP-complete
Nonexistence of a global assignment	coNP-complete
Counting global assignments	#P-complete
Minimum-cardinality fibre obstruction	NP-hard
Minimum-cardinality sensor repair (listed candidates)	NP-complete
Minimum-cardinality unrestricted synthesis (fixed value alphabet)	NP-complete

The hardness statements can be made under nonempty-local-relation promises, so the obstruction is global compatibility rather than a locally empty factor. These are not claims about arbitrary succinct generators. Circuits, automata, and finite-element preprocessors change the input representation and require separate analysis. Fixed-template dichotomy theorems, ETH (exponential-time hypothesis) lower bounds, and parameterised hardness results should be cited only with their exact source hypotheses and parameters; they are not universal classifications of finite ECD presentations. Appendix B records the encodings and the standard reductions behind the table in the verifier-facing input format, and cites the dichotomy, ETH, and W-hierarchy sources together with their exact hypotheses [23, 27, 14, 10]. The design side of sensor repair is established directly rather than transferred: deciding whether a listed candidate family contains a repairing subfamily of at most k sensors is NP-complete, and the reduction preserves the budget, so the problem is W[2]-hard under the parameterisation by the number of added sensors (Proposition 7.3 and Appendix B). The unrestricted formulation, with sensors not drawn from a declared list, is degenerate under free value alphabets and reduces exactly to per-fibre transversal bounds under a fixed value alphabet, with the same NP-completeness and W[2]-hardness in the sensor count (Proposition 7.4).

The width boundary is identification-relative. The primal graph of a finite ECD presentation $\mathcal{P}$ has vertex set X , with an edge between variables that occur together in a scope.

Proposition 5.1 (Identification can destroy low width). *For every finite graph G there is a finite ECD presentation $\mathcal{P}$ and an identification of $\mathcal{P}$ — a partition of its variables into blocks over a common domain, each block merged into one variable and each table replaced by its image under the merge — such that the primal graph of $\mathcal{P}$ is a disjoint union of edges and the primal graph of the quotient presentation is G .*

Proof. For each edge uv of G , take private variables u_{uv} and v_{uv} over a common domain D and the full table $D \times D$ on $\{u_{uv}, v_{uv}\}$. The scopes are pairwise disjoint, so the primal graph is the matching of private copies. Identify all copies of each vertex of G ; every scope becomes an edge $\{u, v\}$ of G , and the full tables merge to full tables, so the quotient presentation has primal graph G . $\square$

Remark 5.2 (Width is measured after identification). The construction is the general form of the observation-variant convention of Section 11: the **split** variant keeps command copies private, while the **same** variant identifies them, and the recorded dense families realise the identified side, their scopes covering the full actuator set, so the primal graph is complete. Width governs decomposition only after the identifications the interface enforces; neither low mechanical-mesh width nor low width of a presentation of separate scenario copies is sufficient by itself. With a three-element domain and disequality tables, the quotient is the graph-colouring instance of G , NP-complete in general [12], while the unmerged presentation is a satisfiable matching of disequalities: width one before identification, the width of G after it.

6 Width-based certifying dynamic programming

Tree decompositions and their dynamic programs are classical [22, 4, 5, 2, 8]; the acyclic case is the join-tree regime of Yannakakis [26]. We fix the certificate-facing formulation. Let $(T, (B_t)_{t \in T})$ be a tree decomposition of the primal graph in which every variable occurs in at least one bag (a variable occurring in no factor can be placed in the root bag). The solver orients T at a root r , assigns each factor to a bag containing its scope, and computes message tables from leaves to root. For a node t , write T_t for the subtree rooted at t , $V_t = \bigcup_{u \in T_t} B_u$ for its variables, $\Sigma_t = B_t \cap B_{\text{pa}(t)}$ for its separator toward the parent (with $\Sigma_r = \emptyset$), and F_t for the factors assigned to t . Value costs

κ are charged at the bag of each variable closest to the root; the running-intersection property makes the bags containing a variable connected, so the charging bag is well defined, and we write $\kappa_t(b) = \sum_{x \text{ charged at } t} \kappa_x(b_x)$.

Lemma 6.1 (Private variables and factor separation). *If $x \in V_t \setminus \Sigma_t$, then x occurs only in bags of T_t . Consequently, every factor whose scope meets $V_t \setminus \Sigma_t$ is assigned to a bag inside T_t .*

Proof. The bags containing x induce a connected subgraph of T , and it meets T_t because $x \in V_t$. If it also met the complement of T_t , a path between the two bags would cross the edge $(t, \text{pa}(t))$, so both B_t and $B_{\text{pa}(t)}$ would contain x , giving $x \in \Sigma_t$ — a contradiction. For the second claim, let factor i be assigned to a bag B_u with $u \notin T_t$ and suppose $x \in S_i \cap (V_t \setminus \Sigma_t)$. Factor containment gives $x \in S_i \subseteq B_u$, so x occurs in a bag outside T_t , contradicting the first claim. $\square$

The message of a node t is the table ν_t , indexed by the assignments s of the separator Σ_t , with entries in $\mathbb{Q}_{\geq 0} \cup \{\infty\}$ in cost mode and feasibility bits in feasibility mode:

$$\nu_t(s) = \min \left\{ \sum_{x \in V_t \setminus \Sigma_t} \kappa_x(a_x) : a \in \prod_{x \in V_t} D_x, a|_{\Sigma_t} = s, a|_{S_i} \in R_i \text{ for every factor } i \text{ assigned in } T_t \right\},$$

the minimum over the empty set being ∞ ; in feasibility mode the entry records whether the set is nonempty. The cost range is justified by Lemma 6.1: the variables of $V_t \setminus \Sigma_t$ occur only in bags of T_t , so their charging bags lie in T_t , while the variables of Σ_t are charged at the parent or above and are settled outside the message. Messages are computed from leaves to the root by the recurrence

$$\nu_t(s) = \min_b \left(\kappa_t(b) + \sum_{c \in \text{ch}(t)} \nu_c(b|_{\Sigma_c}) \right),$$

where the minimum ranges over the assignments b of B_t with $b|_{\Sigma_t} = s$ and $b|_{S_i} \in R_i$ for all $i \in F_t$; since $\Sigma_c = B_c \cap B_t \subseteq B_t$, the restriction $b|_{\Sigma_c}$ is well defined, and for a leaf the sum over children is empty.

Lemma 6.2 (Recurrence correctness). *The recurrence computes, for every node t , the message ν_t defined above.*

Proof. By induction on the size of T_t . Fix an assignment s of Σ_t . An assignment a of V_t extending s decomposes uniquely into its restriction b to B_t and its restrictions to the private parts $V_c \setminus \Sigma_c$ of the child subtrees, and these pieces recombine to a unique a : the sets B_t and $V_c \setminus \Sigma_c$ partition V_t , where a variable of $V_c \setminus \Sigma_c$ cannot lie in B_t without forcing membership in Σ_c by connectedness, and sibling private parts are disjoint by the same argument. By Lemma 6.1, a factor assigned within T_t is either assigned at t , in which case its scope is contained in B_t and it constrains only b , or it is assigned inside some child subtree, in which case it is absorbed into that child's message. The value costs split according to the charging bags. The pieces are coupled only through the separators Σ_c , on which b prescribes values. Hence the minimum over a factors into the displayed minimum over b . $\square$

For heterogeneous domains, the relevant state-space parameter is the weighted width

$$\lambda(T) = \max_{t \in T} \sum_{x \in B_t} \lceil \log_2 |D_x| \rceil.$$

Every message table is indexed by assignments of a separator $\Sigma_t \subseteq B_t$, so it has at most $2^{\lambda(T)}$ entries, since $\prod_{x \in B_t} |D_x| \leq 2^{\lambda(T)}$. Thus the explicit-table recurrences run in $\text{poly}(|I|, |T|) 2^{\lambda(T)}$, up to indexing and arithmetic factors.

Proposition 6.3 (DP certificate soundness). *If an independent verifier recomputes all message tables of a DP-REFUTATION certificate and obtains an infeasible root message, then $\Gamma(\mathcal{P}) = \emptyset$. If it recomputes a DP-OPTIMUM certificate with root value $c < \infty$, then there is a global assignment of value cost c , and no assignment in $\Gamma(\mathcal{P})$ has lower value cost for the declared finite instance.*

Proof. The root has an empty separator, so by Lemma 6.2 the root message aggregates over all assignments of $V_r = X$ subject to every factor — exactly the global policy space:

$$\nu_r() = \min_{a \in \Gamma(\mathcal{P})} \kappa(a).$$

An infeasible root message therefore certifies $\Gamma(\mathcal{P}) = \emptyset$, and a root value $c < \infty$ certifies both that $\Gamma(\mathcal{P}) \neq \emptyset$ and that no global assignment has value cost below c . The verifier recomputes every message table from the decomposition and the instance data alone, so agreement at the root proves the claim for the declared instance. $\square$

Remark 6.4 (A fixed-parameter reading). The bound $\text{poly}(|I|, |T|)2^{\lambda(T)}$ is a fixed-parameter tractability bound [10]: for the input consisting of the instance together with its declared decomposition, both solving and independently recomputing a DP-REFUTATION or DP-OPTIMUM certificate (Proposition 6.3) run in time exponential only in the weighted width λ . The parameterised hardness of the width programme sits on the design side instead: choosing a repairing sensor family is W[2]-hard in the number of added sensors, already at width zero (Proposition 7.3), and unrestricted synthesis under a fixed value alphabet is W[2]-hard in the sensor count (Proposition 7.4), so the design-side barrier is not an artifact of the candidate list. Evaluation width and design budget are separate parameters, and the tractability boundary of the width programme runs between them.

Example 6.5 (A hand-verifiable dynamic program). The recurrence, the charging rule, and both dynamic-programming certificate types can be checked by hand on a four-variable instance. Let $X = \{x_1, x_2, x_3, x_4\}$ with $D_x = \{0, 1\}$ for every x , value costs $\kappa_x(1) = 1$ and $\kappa_x(0) = 0$ as in the benchmark convention of Remark 2.6, and three pair factors

$$\begin{aligned} R_1 &= \{(0, 1), (1, 0), (1, 1)\} \text{ on } S_1 = \{x_1, x_2\}, & R_2 &= \{(0, 0), (1, 1)\} \text{ on } S_2 = \{x_2, x_3\}, \\ R_3 &= \{(0, 1), (1, 0)\} \text{ on } S_3 = \{x_3, x_4\}, \end{aligned}$$

read as *at least one*, *equality*, and *exactly one*. The primal graph is the path $x_1 - x_2 - x_3 - x_4$. A valid tree decomposition is the path of bags

$$B_1 = \{x_1, x_2\}, \quad B_2 = \{x_2, x_3\}, \quad B_3 = \{x_3, x_4\},$$

rooted at $r = B_3$, with factor i assigned to bag B_i . The charging rule charges x_1 at B_1 , x_2 at B_2 (its bag closest to the root), and x_3, x_4 at B_3 ; the separators are $\Sigma_1 = \{x_2\}$, $\Sigma_2 = \{x_3\}$, and $\Sigma_r = \emptyset$. The instance is small enough to check entirely by hand; it is a pedagogical check of the recurrence and the certificate reading, not a recorded artifact run.

The leaf message ν_1 , indexed by the assignments of $\Sigma_1 = \{x_2\}$, is

$$\nu_1(x_2=0) = 1, \quad \nu_1(x_2=1) = 0 :$$

when $x_2 = 0$ the factor R_1 admits only the bag assignment $(1, 0)$, whose charge is $\kappa_{x_1}(1) = 1$, and when $x_2 = 1$ it admits $(0, 1)$ and $(1, 1)$, the cheaper being $\kappa_{x_1}(0) = 0$; the sum over children is empty at a leaf. The message ν_2 , indexed by $\Sigma_2 = \{x_3\}$, adds the charge of x_2 , the only variable charged at B_2 :

$$\nu_2(x_3=0) = \kappa_{x_2}(0) + \nu_1(x_2=0) = 0 + 1 = 1, \quad \nu_2(x_3=1) = \kappa_{x_2}(1) + \nu_1(x_2=1) = 1 + 0 = 1,$$

since R_2 forces $x_2 = x_3$. At the root, R_3 admits exactly the bag assignments $(0, 1)$ and $(1, 0)$, both charged at B_3 :

$$\nu_r() = \min\{\kappa_{x_3}(0) + \kappa_{x_4}(1) + \nu_2(x_3=0), \kappa_{x_3}(1) + \kappa_{x_4}(0) + \nu_2(x_3=1)\} = 2.$$

Backtracking through the minimising entries yields the two optimal sections $(1, 0, 0, 1)$ and $(0, 1, 1, 0)$; a brute-force sweep over the sixteen assignments confirms that the global sections are exactly these two and $(1, 1, 1, 0)$, of value costs 2, 2, and 3. Along the root path each variable is charged exactly once — x_1 at B_1 , x_2 at B_2 , x_3 and x_4 at B_3 , totalling $1 + 0 + 0 + 1 = 2$ — which is the non-double-counting guaranteed by Lemma 6.1.

The certificate reading is immediate. The decomposition with its bags, edges, root, and factor placement is a **TREE-DECOMP** certificate; together with the three message tables and the root value 2 it constitutes a **DP-OPTIMUM** certificate of value 2, and an independent verifier recomputes ν_1 , ν_2 , and ν_r from the instance and the decomposition alone, as in Proposition 6.3. In feasibility mode the same tables carry bits, and the root bit records satisfiability. Replacing the three tables by

$$R'_1 = \{(0, 0), (1, 0)\}, \quad R'_2 = \{(0, 1)\}, \quad R'_3 = \{(0, 0), (0, 1)\}$$

gives $\nu'_1 = (0, \infty)$ and $\nu'_2 = (\infty, 0)$ (entries ordered by the separator value 0, 1) and an infeasible root, since R'_3 forces $x_3 = 0$ while $\nu'_2(x_3=0) = \infty$; the same three tables are then a **DP-REFUTATION** certificate, and indeed R'_1 forces $x_2 = 0$, R'_2 then forces $x_3 = 1$, and R'_3 forces $x_3 = 0$. Finally, $\lambda(T) = 2$ here, since each bag holds two Boolean variables, so every message table has at most $2^{\lambda(T)} = 4$ entries; the separators are single variables, so the displayed tables have two entries each.

7 Sensor repair

A sensor architecture is part of the presentation. Fix one-shot data $W, o, A, (S_w)$ as in Section 4. A conflict core is a set $C \subseteq o^{-1}(z)$ of scenarios within a single observation fibre such that $\bigcap_{w \in C} S_w = \emptyset$; it is inclusion-minimal when no proper subset is a conflict core. For a core C , let

$$E_C = \{s : s \text{ is nonconstant on } C\}$$

be the candidate sensors that separate scenarios of the core. Adding a sensor family T refines the observation map to $w \mapsto (o(w), (s(w))_{s \in T})$; the family repairs the one-shot problem when the refined problem admits an observation-uniform selector.

Theorem 7.1 (One-shot sensor repair).

1. A sensor family T repairs the one-shot problem if and only if

$$T \cap E_C \neq \emptyset$$

for every inclusion-minimal conflict core C .

2. For any listed family of conflict cores, hitting every listed core is necessary for repair, and it is sufficient — hence exact — when the list contains all inclusion-minimal conflict cores.

Proof. Necessity in (1). Suppose $T \cap E_C = \emptyset$ for a conflict core $C \subseteq o^{-1}(z)$. Then every sensor $s \in T$ is constant on C , so all scenarios of C share one refined observation value (z, v) . The refined fibre over (z, v) contains C , and its admissible intersection is contained in $\bigcap_{w \in C} S_w = \emptyset$. By Theorem 4.1 the refined problem admits no selector, so T does not repair.

Sufficiency in (1). Suppose some refined fibre F' fails, that is, $\bigcap_{w \in F'} S_w = \emptyset$. All scenarios of F' share the original observation value z , so F' is itself a conflict core of the original problem and, being finite and nonempty, contains an inclusion-minimal core $C_{\min}$. Every sensor $s \in T$ is constant on F' , hence on $C_{\min}$, so $T \cap E_{C_{\min}} = \emptyset$, contradicting the hypothesis. Hence every refined fibre has nonempty admissible intersection, and Theorem 4.1 yields a selector for the refined problem.

Claim (2). Necessity holds because a repairing family hits every core, by the contrapositive of the necessity direction in (1). When the list contains all inclusion-minimal cores, hitting the list implies hitting every inclusion-minimal core, which repairs by the sufficiency direction in (1); conversely, a repairing family hits every core, in particular every listed one. Under this hypothesis the hitting-set condition is therefore exact for one-shot repair. $\square$

Remark 7.2 (Minimal cores, not failing fibres). Hitting the sensors nonconstant on whole failing fibres is not sufficient for repair. A fibre may fail through a small core while the selected sensors separate only scenarios outside that core; the refined fibre containing the core then still fails. Minimal failing fibres and inclusion-minimal conflict cores differ in general: a minimal core need not exhaust its fibre, and a failing fibre need not be minimal. The exact objects for one-shot repair are the inclusion-minimal conflict cores.

The theorem separates two tasks: synthesising a policy for a fixed observation architecture, and designing an observation architecture that makes policy synthesis possible. The second task contains hitting-set structure even when the first task is easy.

Proposition 7.3 (Sensor-design hardness). *Deciding whether a listed family of candidate sensors contains a subfamily of at most k sensors that repairs the one-shot problem is NP-complete, and W[2]-hard under the parameterisation by k . Both statements hold already for instances with one observation fibre and two actions, whose single-channel presentations have one unary factor per scenario and an edgeless primal graph.*

Proof. The decision problem lies in NP: a witness is the chosen subfamily together with a selector for the refined problem, and a scan linear in the scenario list checks the selector against every scenario. For hardness, the set-cover construction of Appendix B maps a covering instance with budget k to a one-shot instance with one observation fibre and two actions in which the inclusion-minimal conflict cores are exactly the pairs $\{w_0, w_j\}$ and a candidate sensor separates w_0 from w_j exactly when its covering set contains j ; by Theorem 7.1, the repairing subfamilies of size at most k are exactly the covers of size at most k . The construction preserves the budget, and set cover is W[2]-complete under the standard parameterisation [10], so the NP-hardness and the W[2]-hardness both follow. The single-channel presentation of the constructed instance has one variable per observation value — one, here — and one unary factor per scenario, an edgeless primal graph: the hardness is present at width zero, and no decomposition of the declared instance removes it (Remark 6.4). $\square$

The unrestricted formulation — sensors not drawn from a declared candidate list — admits a complete classification, which also identifies what the candidate list contributes.

Proposition 7.4 (Unrestricted sensor synthesis). *Fix one-shot data $W, o, A, (S_w)$ with $S_w \neq \emptyset$ for every scenario w .*

1. *Under unrestricted sensor value sets, minimum-cardinality synthesis is degenerate: the empty family repairs when the one-shot problem already admits a selector, and otherwise a single sensor — one taking distinct values on the scenarios of each observation fibre — repairs, so the minimum is 0 or 1.*

2. For a fixed value alphabet of size $a \geq 2$, a family of m sensors repairs if and only if, for every realised observation value z , the admissible family $(S_w)_{w \in o^{-1}(z)}$ admits a transversal — a set of actions meeting every S_w — of cardinality at most a^m . The minimum sensor count is therefore $\max_z \lceil \log_a \tau_z \rceil$, where τ_z is the transversal number of the fibre family and the maximum is over realised observation values.
3. Deciding whether m sensors with values in a fixed alphabet of size $a \geq 2$ exist is NP-complete and W[2]-hard under the parameterisation by m , already with one observation fibre and $a = 2$.

Proof. (1) The empty family is the refining family of the original problem, which admits a selector exactly under the fibre criterion of Theorem 4.1. Otherwise some fibre fails. A sensor taking distinct values on the scenarios of each fibre is nonconstant on every inclusion-minimal conflict core: every core lies in one fibre and has at least two scenarios, since $S_w \neq \emptyset$ rules out singleton cores. By Theorem 7.1 the singleton family repairs, and no nonempty family has fewer members.

(2) Let T be a repairing family $s_1, \dots, s_m$ with values in the fixed a -element set. Within one observation fibre $o^{-1}(z)$, the nonempty refined fibres partition the scenarios into at most a^m classes, each with nonempty admissible intersection; one action chosen from each nonempty class yields a transversal of cardinality at most a^m , since every scenario's admissible set contains the action of its class. Conversely, suppose each fibre family admits a transversal T_z with $|T_z| \leq a^m$. Fix an order on each T_z , assign every $w \in o^{-1}(z)$ an element $t(w) \in T_z \cap S_w$, and encode the rank of $t(w)$ in m digits to base a ; the j -th digits of the codes define s_j . Scenarios of one fibre assigned the same element receive identical codes, and scenarios assigned different elements receive different ones, so each nonempty refined fibre consists exactly of the scenarios of one fibre assigned to one element, whose admissible intersection contains that element. The displayed minimum follows by taking logarithms; the empty family is the case $a^0 = 1$, which holds exactly when every fibre family has a one-element transversal — the fibre criterion itself.

(3) Membership in NP is witnessed by the m sensor tables together with a selector for the refined problem, checkable by a scan over the scenario list. Hardness, both the NP-completeness and the W[2]-hardness in the sensor count, is the transversal padding construction of Appendix B, which reduces hitting set to the decision problem with a single observation fibre and $a = 2$. $\square$

8 Certified affine mechanics front end

The mechanics front end supplies finite relations; the finite solver certifies policies or refutations for those relations. For affine response rows,

$$y_i(a) = d_i + \sum_{x \in S_i} g_{ix} a_x,$$

exact arithmetic or certified intervals classify each row as safe, unsafe, or unresolved. More generally, for a symmetric positive-definite discrete solve $Ku = f$, residual $r = f - K\hat{u}$, and certified lower spectral bound $\underline{\lambda} \leq \lambda_{\min}(K)$, one has

$$\|u - \hat{u}\|_2 \leq \frac{\|r\|_2}{\underline{\lambda}}, \quad |c^T(u - \hat{u})| \leq \|c\|_2 \frac{\|r\|_2}{\underline{\lambda}}.$$

Rows whose certified response intervals cross a threshold remain unresolved.

Proposition 8.1 (Exact affine row certificates). *For a finite affine response model with rational coefficients, finite actuator domains, and a rational safety interval, exact rational evaluation produces FE-ROW certificates whose safe rows form the corresponding finite ECD relation table. The verifier can recompute every row exactly.*

Proof. For each action tuple, the response is a rational number obtained by a finite sum of rational products. The verifier recomputes this value, compares the singleton response interval with the declared safety interval, and accepts the row classification exactly when the comparison agrees. The finite relation for a mechanical scenario is precisely the set of action tuples whose certificates verify as safe. $\square$

Proposition 8.2 (Inner–outer finite soundness). *Let P^- , P^{true} , and P^+ be presentations sharing variables, domains, and factor scopes. If finite row certificates give*

$$R_i^- \subseteq R_i^{\text{true}} \subseteq R_i^+$$

for every factor, then

$$\Gamma(P^-) \subseteq \Gamma(P^{\text{true}}) \subseteq \Gamma(P^+).$$

Thus inner satisfiability gives a certified true finite-model policy, and outer unsatisfiability gives a certified true finite-model refutation. Inner unsatisfiability with outer satisfiability is inconclusive.

Proof. Let $a \in \Gamma(P^-)$. For every factor i , $a|_{S_i} \in R_i^- \subseteq R_i^{\text{true}}$, so $a \in \Gamma(P^{\text{true}})$. The inclusion $\Gamma(P^{\text{true}}) \subseteq \Gamma(P^+)$ follows identically from $R_i^{\text{true}} \subseteq R_i^+$. Consequently, an assignment certified by the inner model satisfies the true relations, so inner satisfiability exhibits a true finite-model policy; if the outer model has no global assignment, then neither has the true model, so outer unsatisfiability refutes it; and when the inner model is unsatisfiable while the outer model is satisfiable, both alternatives for the true model remain open, which is recorded as inconclusive. $\square$

8.1 Declared structural models and certified influence rows

The affine models above can themselves be certified from declared structural data. A declared structural model consists of nodes with rational coordinates and support flags, members with declared rational axial stiffnesses, actuation patterns (node forces, or self-equilibrated axial member pairs modelling pre-tension), load scenarios, and monitored linear functionals (member forces or displacement components). Members use rational directions — axis-aligned chords and 3-4-5 diagonals — so every direction cosine is rational and the assembled free-degree stiffness matrix K is exact. Two model kinds cover the families below: pin-jointed planar trusses and grounded scalar mass-spring chains.

Proposition 8.3 (Structural row certificates). *For a declared structural model with rational data, exact assembly and exact rational solves produce, for every scenario and monitor, an affine row*

$$y_i(a) = d_i + \sum_{x \in S_i} g_{ix} a_x$$

with rational d_i and g_{ix} , together with a certificate whose base-state solves are checkable by exact residual substitution $K\hat{u} = f$ against an independently assembled stiffness matrix, and whose row data are checkable by regeneration. The derived relation table is exactly the set of actuator tuples whose row responses lie in the declared safe interval.

Proof. Assembly accumulates rational dyads, so K is rational. Exact elimination over the rationals returns the unique solution of $Ku = f$, and the verifier proves it by substitution: $K\hat{u} - f = 0$ exactly. Monitors are rational linear functionals of u , so d_i and every g_{ix} are rational and recomputable, and the classification compares the exact response with the declared rational interval. $\square$

One structural distinction drives the benchmark families. For the grounded serial chain, the exact influence of a pre-tension pair on a spring force is local: the pair changes only its own spring's force, because the pattern is self-equilibrated and the structure is statically determinate. Node forces have dense influence on displacement and internal-force monitors, and in statically indeterminate lattices even pre-tension pairs influence every member force. Locality of exact influence, not sparsity of an approximate solver, is what the measured scopes record.

9 Width-adaptive sparsification

Retaining only part of each row's scope trades interval uncertainty for width; this section proves that the resulting refinement is monotone and certifiable. For affine rows, retain $J_i \subseteq S_i$ and bound omitted terms by

$$\underline{\eta}_i(J) = \sum_{x \in S_i \setminus J_i} \min_{v \in D_x} g_{ix}v, \quad \bar{\eta}_i(J) = \sum_{x \in S_i \setminus J_i} \max_{v \in D_x} g_{ix}v.$$

All extrema in this section are over the finite numeric domains, as in all instances of this article. The inner sparse relation keeps a retained assignment only if every completion of omitted variables is certified safe. The outer sparse relation keeps it if some completion is not certified unsafe. Write Γ_J^- and Γ_J^+ for the global policy spaces of the inner and outer sparse instances: the assignments in $\prod_{x \in X} D_x$ whose restrictions to the retained scopes J_i lie in the corresponding sparse relations. Write further Γ^{true} for $\Gamma(P^{\text{true}})$, the global policy space of the true, unsparisified instance.

Theorem 9.1 (Sparsification ladder). *If retained scopes grow from J_i to J'_i , then*

$$\Gamma_J^- \subseteq \Gamma_{J'}^- \subseteq \Gamma^{\text{true}} \subseteq \Gamma_{J'}^+ \subseteq \Gamma_J^+.$$

Proof. All five policy spaces live in $\prod_{x \in X} D_x$; we compare them assignment by assignment. Fix a and a factor i .

First, $a \in \Gamma_J^-$ implies $a \in \Gamma_{J'}^-$. The coarse inner hypothesis says that every completion of $a|_{J_i}$ to S_i is certified safe. The completions of $a|_{J'_i}$ to S_i form a subset of these: they fix the newly retained variables $J'_i \setminus J_i$ to the values of a , which lie in the respective domains. Hence every fine completion is certified safe, which is the fine inner condition.

Second, $a \in \Gamma_{J'}^-$ implies $a \in \Gamma^{\text{true}}$: applying the fine inner hypothesis to the completion given by a itself, the response of $a|_{S_i}$ is certified safe, so $a|_{S_i} \in R_i^{\text{true}}$.

Third, $a \in \Gamma^{\text{true}}$ implies $a \in \Gamma_{J'}^+$: the true row $a|_{S_i}$ is safe, so $a|_{J'_i}$ has a completion that is not certified unsafe, namely a itself.

Fourth, $a \in \Gamma_{J'}^+$ implies $a \in \Gamma_J^+$. The fine outer hypothesis provides a completion v of $a|_{J'_i}$ to S_i whose response is not certified unsafe. Extend it to a completion of $a|_{J_i}$ by assigning each $x \in J'_i \setminus J_i$ the value a_x . The resulting completion agrees with a on J'_i and with v on $S_i \setminus J'_i$, so its response equals that of the fine witness and is not certified unsafe; hence $a|_{J_i}$ is a coarse outer row.

In interval terms, the same steps say that for a fine retained assignment a' on J'_i with $a'|_{J_i} = a$, the conditioned response interval

$$d_i + \sum_{x \in J'_i} g_{ix}a'_x + [\underline{\eta}_i(J'), \bar{\eta}_i(J')] \subseteq d_i + \sum_{x \in J_i} g_{ix}a_x + [\underline{\eta}_i(J), \bar{\eta}_i(J)],$$

because each newly retained value $a'_x \in D_x$ lies in the per-term range $[\min_{v \in D_x} g_{ix}v, \max_{v \in D_x} g_{ix}v]$; inner safety transfers to subintervals and outer intersection to superintervals. $\square$

This is a soundness theorem, not a performance theorem. It says that refinement is monotone and certifiable; whether it improves runtime on a class of mechanics instances is an empirical question.

Table 1 records the certificate-bearing outcomes at a fixed profile. Both certification rows are stable under refinement $J \subseteq J'$: a certified satisfiable pair stays satisfiable because $\Gamma_{J'}^- \supseteq \Gamma_J^-$, and a certified unsatisfiable pair stays unsatisfiable because $\Gamma_{J'}^+ \subseteq \Gamma_J^+$; only an inconclusive pair can move, which is the operational content of the monotonicity claim.

Configuration at profile J	Certificates on the sparse instances	Status for Γ^{true}
$\Gamma_J^- \neq \emptyset$	POLICY (inner) + SPARSIFICATION-BATCH	certified satisfiable: every $a \in \Gamma_J^-$ is a true-model policy
$\Gamma_J^+ = \emptyset$	DP-REFUTATION (outer) + SPARSIFICATION-BATCH	certified unsatisfiable: $\Gamma^{\text{true}} = \emptyset$
$\Gamma_J^- = \emptyset$ and $\Gamma_J^+ \neq \emptyset$	DP-REFUTATION (inner), POLICY (outer), + SPARSIFICATION-BATCH	inconclusive for the true model at profile J

Table 1: Certificate-bearing outcomes of the sparsification ladder at a fixed retained profile J . The SPARSIFICATION-BATCH certificate proves the bracket $\Gamma_J^- \subseteq \Gamma^{\text{true}} \subseteq \Gamma_J^+$; the two one-sided rows transport claims through the bracket, and the third row is the certified inconclusive case. The recorded study of Section 12 reads every rung of every ladder through these rows and additionally re-verifies inner policies directly against the full certified rows.

When value costs are present, the same bracket gives certified cost bounds

$$\min_{a \in \Gamma_J^+} \kappa(a) \leq \min_{a \in \Gamma^{\text{true}}} \kappa(a) \leq \min_{a \in \Gamma_J^-} \kappa(a), \quad \min \emptyset = \infty,$$

the lower bound certified by a DP-OPTIMUM run on the outer instance and the upper bound by one on the inner instance; the recorded study of Section 12 classifies brackets and does not tabulate these cost bounds.

10 Exact demonstrator: two springs and one repairing sensor

The smallest mechanics example is already enough to show why observational uniformity matters; it assembles the certificate layers of the preceding sections, while the sparsification ladder of Section 9 is exercised by the study of Section 12. Let

$$u(f, a_1, a_2) = \frac{f - a_1 - a_2}{2}, \quad f \in \{1, 2\}, \quad a_1, a_2 \in \{0, 1\},$$

and impose the safety condition $|u| \leq 0.25$. Exact evaluation gives

$$R_1 = \{(0, 1), (1, 0)\}, \quad R_2 = \{(1, 1)\}.$$

Thus each load scenario has an admissible command, but if the two loads have the same observation, no observation-uniform command can satisfy both relations. The common observation fibre is locally conflicting: the fibre itself is the unique inclusion-minimal conflict core, since $R_1 \cap R_2 = \emptyset$ while each relation alone is nonempty. A threshold sensor separating the two loads is therefore a repairing sensor in the sense of Theorem 7.1: it separates the scenarios of the minimal core. It creates two information states; the repaired finite presentation is satisfiable, for example by choosing $(0, 1)$ or $(1, 0)$ in the low-load state and $(1, 1)$ in the high-load state.

The physical reading deserves emphasis, because it mirrors the physical reading the foundational monograph gives its Bell obstruction: the obstruction has an operational face as a repair — here, the acquisition of a sensor. A controller commanded without load sensing must remain admissible for every load that its sensors fail to distinguish; here the two loads share one information state. What the refutation certificate certifies is therefore not that the structure is unsafe but that the declared information structure is insufficient: the obstruction sits in the observation map, not in the mechanics. The repairing sensor is an instrumentation decision with a certified justification — Theorem 7.1 guarantees that separating the core’s scenarios restores satisfiability, and the brute-force repair certificate on the ten-bar instance of Section 11 exhibits the same investment at structural scale. Figure 1(a) sketches the two-spring arrangement.

The example is a correctness demonstrator rather than a benchmark. It exercises exact affine row generation, fibre-conflict certificates, sensor repair, positive policy certificates, negative refutation certificates, and corrupted certificate rejection (gate records in Section 13).

11 Mechanics benchmark suite

The computational study is a certification study over declared structural models, not a speed test. Four families (Table 2) cover the structural distinctions above; each declares two load scenarios, Boolean actuator domains, and a rational safe interval, and each is instantiated in the two observation variants of the two-spring convention: **same** (both scenarios share one command instance, so an observation-uniform policy must satisfy both) and **split** (a separating sensor gives each scenario its own command copy). The **same** variant is the identification of the **split** variant’s command copies; Proposition 5.1 and Remark 5.2 record what identification does to width. Decompositions are constructed by a deterministic minimum-fill elimination heuristic [13, 18] and validated by the tree-decomposition verifier; instances carry per-actuator value costs (Remark 2.6), so optima report the minimum number of active actuators. Figure 1 sketches the demonstrator and the declared structure of the four families; the sketches follow the artifact’s model generators.

Family	Structure	Actuators	Monitors	Influence
ten-bar	cantilever, 3×4 bays	vertical node forces	member forces	dense (width 3)
chain-pretension	grounded chain, unit springs	pair per spring	spring forces	local (width 0)
dense-chain	grounded chain, stiffnesses 1, 2	node force per node	mid-chain displ.	dense ($n - 1$)
lattice	cross-braced two-row lattice	pairs on vert. and diag.	member forces	dense (grows)

Table 2: Mechanics benchmark families. Lattice width grows as $6C - 1$ in the column count C ; relation tables beyond the enumeration limit are marked explicitly.

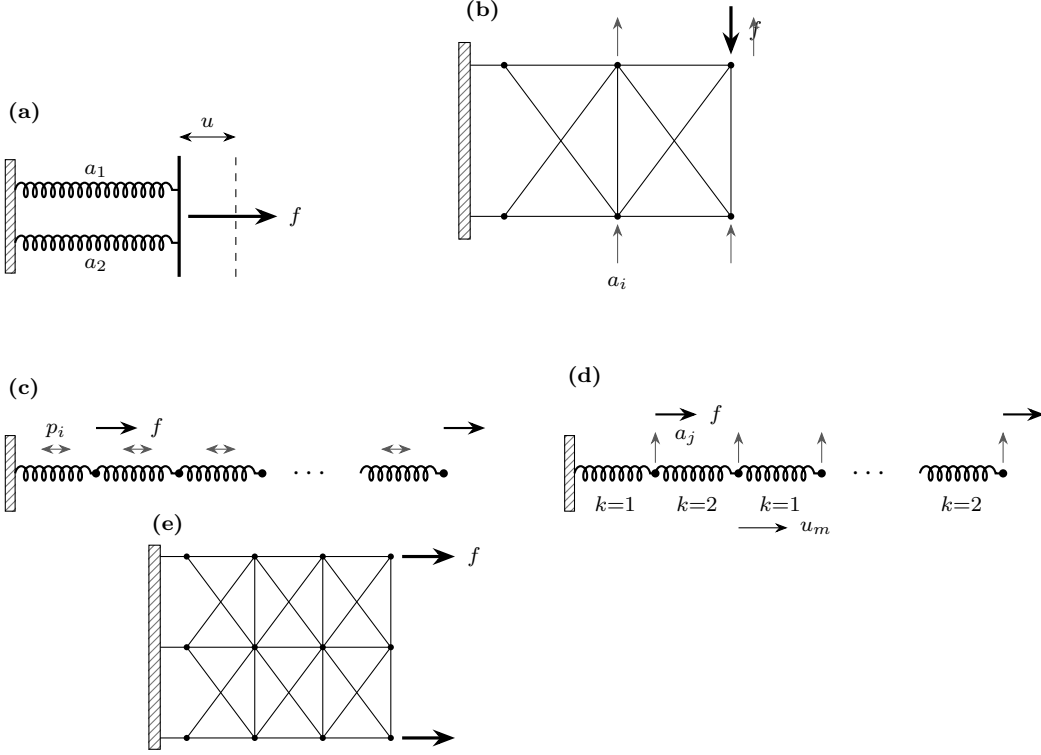


Figure 1: Schematic of the demonstrator and the four benchmark families, drawn after the artifact’s model generators (not to scale). (a) Two-spring demonstrator of Section 10: two unit springs in parallel anchor a rigid bar to the wall; the load f and the two actuator commands a_1, a_2 determine the bar displacement $u = (f - a_1 - a_2)/2$ (displaced position dashed). (b) Ten-bar family: the classic six-node cantilever with two 3×4 bays and 3–4–5 diagonals; grey arrows mark the four vertical node-force actuators at the free nodes, the black arrow the quantised tip loads, and member forces are monitored. (c) Chain-pretension family: unit springs grounded at one end, each carrying a self-equilibrated pre-tension pair (double arrows, p_i); spring forces are monitored and the two load scenarios act at the chain ends. (d) Dense-chain family: alternating spring stiffnesses 1 and 2, one node-force actuator per node (grey arrows), a monitored mid-chain displacement u_m , and end loads. (e) Lattice family: the cross-braced lattice with two bay-rows and C columns (three shown), fully supported at the left edge; pre-tension pairs act on the verticals and diagonals, member forces are monitored, and the two quantised loads act at the free-edge corners. Table 2 records the corresponding influence structure and widths.

Table 3 reports the recorded gate: all twenty row batches are independently verified, all eighteen constructed decompositions are validated, all eighteen dynamic-programming certificates are re-verified, each constructed split instance is solved with a verified optimum policy and each constructed same instance is refuted with a verified DP-REFUTATION, while the two remaining cases, the largest lattice instances, carry certified influence scopes and recorded scope widths but no constructible instance: their relation tables exceed the enumeration cap and are marked **RELATION-TABLE-LIMIT**. Corrupted row batches and corrupted dynamic-programming certificates are rejected on the small representatives. The ten-bar family is cross-validated against the exact affine row generator of Section 8: the derived finite instances agree. Its same-observation instance carries a brute-force sensor-repair certificate selecting one threshold sensor with certified optimality, so the quantised-sensor structure of the two-spring demonstrator reappears in a structural model.

Instance	$ X $	rows	width	status	DP (s)	verify (s)
ten-bar, same	4	20	3	UNSAT	0.001	0.001
ten-bar, split	8	20	3	OPT	0.001	0.001
chain $n=8$, same / split	8 / 16	16	0	UNSAT / OPT	0.000 / 0.001	0.000 / 0.001
chain $n=256$, same / split	256 / 512	512	0	UNSAT / OPT	0.011 / 0.031	0.018 / 0.059
chain $n=1024$, same / split	1024 / 2048	2048	0	UNSAT / OPT	0.097 / 0.292	0.159 / 0.548
dense-chain $n=8$, same / split	8 / 16	2	7	UNSAT / OPT	0.006 / 0.012	0.009 / 0.017
dense-chain $n=16$, same / split	16 / 32	2	15	UNSAT / OPT	2.37 / 4.57	3.06 / 6.33
lattice $C=1$, same / split	6 / 12	18	5	UNSAT / OPT	0.002 / 0.003	0.002 / 0.004
lattice $C=2$, same / split	12 / 24	36	11	UNSAT / OPT	0.138 / 0.307	0.217 / 0.328
lattice $C=3$, scopes	—	54	17	RELATION-TABLE-LIMIT	—	—
lattice $C=4$, scopes	—	72	23	RELATION-TABLE-LIMIT	—	—

Table 3: Recorded mechanics benchmark gate (selected sizes; the artifact’s recorded reports contain all twenty cases and timings). Every constructed same-observation instance is refuted with a verified certificate and every constructed split instance solved with a verified optimum policy, including the 1024-spring chain and the width-11 lattice. Status entries abbreviate the four-status discipline of Remark 3.1: UNSAT and OPT record verified DP-REFUTATION and DP-OPTIMUM certificates, that is, CERTIFIED-UNSAT and CERTIFIED-OPT. Lattice column counts three and four have certified influence scopes but relation tables beyond the enumeration limit (RELATION-TABLE-LIMIT), so no instance is constructed for them; for these two rows the artifact’s recorded width field is the influence-scope size (18 and 24 actuators — every certified row’s scope is the full actuator set, so the primal graph is complete), and the tabulated 17 and 23 are the treewidths this forces, continuing the $6C - 1$ growth of the constructed lattices.

Three recorded observations carry structural content. First, the pre-tensioned chain certifies refutations at 1024 springs with width-0 decompositions: the exact locality of self-equilibrated pre-tension makes the finite presentation bounded-width at every scale, and the pipeline cost is dominated by certified relation generation rather than by search. Second, the dense-influence families realise the width boundary: the measured scope of every row equals the full actuator set, so decomposition width grows linearly and the relation tables themselves exceed the enumeration limit at moderate sizes. Third, the sensor structure is decisive at every scale: the same-observation instances are unsatisfiable and the split instances satisfiable in every recorded family case, with the ten-bar repair certificate recorded above exhibiting the minimal separating sensor.

12 Width-adaptive sparsification study

The sparsification ladder of Section 9 is evaluated on retained-scope ladders by structural proximity: the retained scope of a row at radius r keeps the actuators anchored within member-graph distance r of the monitored member or node, and the full radius is the no-sparsification ablation. For every rung the study verifies the sparsification certificate and the pairwise ladder monotonicity with the previous rung, solves the inner and outer instances with verified dynamic-programming certificates, and classifies the bracket through the rows of Table 1: certified satisfiable when an inner policy, extended by zeros on the omitted actuators (every completion of an inner-retained assignment is certified safe, so the zero completion is one), verifies exactly against the full certified rows; certified unsatisfiable when the outer model is refuted, which is sound because outer relations contain the true relations; and inconclusive when the inner model is unsatisfiable while the outer model is satisfiable. Both observation variants are evaluated at every rung.

Family	radius	retained	inner width	unresolved	brackets (same / split)
ten-bar	0	1.6	3	0.50	inconclusive / inconclusive
ten-bar	1–full	4.0	3	0.00	certified / certified
chain $n=64$	0–full	1.0	0	0.00	certified / certified
dense-chain $n=12$	0	1.0	0	1.00	inconclusive / inconclusive
dense-chain $n=12$	1	3.0	2	1.00	inconclusive / inconclusive
dense-chain $n=12$	2	5.0	4	1.00	inconclusive / inconclusive
dense-chain $n=12$	3	7.0	6	1.00	inconclusive / inconclusive
dense-chain $n=12$	full	12.0	11	0.00	certified / certified
lattice $C=1$	0	0.67	0	0.33	inconclusive / inconclusive
lattice $C=1$	1	3.56	5	0.15	inconclusive / inconclusive
lattice $C=1$	2–full	5.56	5	0.00	certified / certified
lattice $C=2$	0	0.67	0	0.64	inconclusive / inconclusive
lattice $C=2$	1	6.0	11	0.27	inconclusive / inconclusive
lattice $C=2$	full	12.0	11	0.00	certified / certified
lattice $C=3$	0	0.67	0	0.70	inconclusive / inconclusive
lattice $C=3$	1	6.67	14	0.34	inconclusive / inconclusive

Table 4: Recorded width-adaptive sparsification study (selected rungs; the artifact’s recorded reports contain the complete ladder data). Retained is the mean retained-scope size; unresolved is the mean share of outer rows that are not inner rows; brackets are the certified or inconclusive classifications of the same- and split-observation variants. The complete study covers twenty rungs — a ten-bar ladder of four radii, a two-rung chain-pretension ablation whose rungs coincide because radius 0 already retains every scope (displayed as the single range 0–full), a dense-chain ladder of five radii, and lattice ladders of four, three, and two radii at column counts one, two, and three — with fourteen verified ladder steps; the compressed radius ranges mark consecutive rungs whose brackets certify, and the recorded values shown are those of the finest rung of each range, whose inner widths agree with the full-instance widths of Table 3. All 20 sparsification certificates and 14 ladder steps verify; the dense instance of lattice $C=3$ is beyond the relation-table enumeration limit (**RELATION-TABLE-LIMIT**).

The recorded outcome of Table 4 is a soundness result with an honest negative component. Monotonicity holds and is certified at every ladder step, the unresolved fraction decreases monotonically with the radius in every family, and the brackets agree with the true statuses at the full radius. But on the calibrated families the brackets certify only at or near the full radius: the omitted-interval tails dominate the declared safety windows at small radii, so aggressive sparsification is sound yet inconclusive, and the retained scopes at intermediate radii overlap enough that the inner width need not drop below the dense width. The study therefore supports the ladder theorem and the bracket discipline, and records that the calibrated models do not yet exhibit a practical width-reduction benefit from proximity-based retention.

13 Prototype artifact and reproducibility

The artifact includes explicit-table solving, tree-decomposition DP certificates, exact affine row certificates, sparsification certificates, controlled-width benchmark generation, baseline encoders for DIMACS CNF, LP-format MILP, and a minimal CP-SAT representation, optional external-solver adapters, a fixed-hardware external-baseline sweep, external-result cross-checking, report

summaries, and SVG plot generation. The mechanics layer adds an exact rational structural front end with independently verified row batches, the four benchmark families with both observation variants, minimum-fill decompositions, the sparsification ladder study, and the comparative scaling study on mechanics-derived instances, together with its version-exact replication and dense-family extension drivers and the forensic audit tools for the recorded reports. The repository-ready entry point is

`make reproduce`

with `./reproduce.sh` as an equivalent shell command. Appendix A documents the certificate formats and the reproduction and archive protocols in detail.

A compact verification table from the verification gate is as follows; the phase labels follow the artifact’s recorded build phases, phase 6 being the recorded mechanics comparative-scaling study.

Gate	Status
Two-spring, same observation	CERTIFIED-UNSAT
Two-spring, split observation	CERTIFIED-SAT
Fibre conflict certificate	VERIFIED
Corrupted fibre certificate	REJECTED
Threshold sensor repair	CERTIFIED-OPT
Tree-decomposition optimum certificate	VERIFIED
Tree-decomposition refutation certificate	VERIFIED
Exact affine FE-ROW batch	VERIFIED
Corrupted FE-ROW batch	REJECTED
Sparsification ladder	VERIFIED
External baseline sweep, five solvers	COMPLETE
External satisfiable-model cross-checks	VERIFIED
External refutation cross-checks	VERIFIED
Mechanics row batches, twenty cases	VERIFIED
Corrupted mechanics row batch	REJECTED
Corrupted mechanics DP certificate	REJECTED
Ten-bar affine cross-validation	VERIFIED
Ten-bar threshold sensor repair	CERTIFIED-OPT
Mechanics sparsification certificates	VERIFIED
Mechanics ladder monotonicity steps	VERIFIED
Mechanics baseline sweep, five solvers	COMPLETE
Phase-6 sweep replication, five solvers	COMPLETE
Dense-extension generation, four cases	VERIFIED
Dense-extension DP memory-limit case	INCONCLUSIVE
Dense-extension sweep, five solvers	COMPLETE
Dense-extension external cross-checks	VERIFIED
Recorded sweep counters vs per-instance records	VERIFIED

Table 5: Recorded verification-gate rows (selected). The mathematical statuses follow the four-status discipline of Remark 3.1: CERTIFIED-SAT, CERTIFIED-UNSAT, and CERTIFIED-OPT transport certified claims, and INCONCLUSIVE records the absence of one. VERIFIED, REJECTED, and COMPLETE are bookkeeping markers for the state of a gate check, not answers.

Controlled-width benchmark families include paths, binary trees, banded chains, and small grids with supplied decompositions, together with an unsatisfiable odd-ring family carrying a supplied

path decomposition of width two. The reports record instance size, ordinary width, weighted width, certificate size, and verification time. These are certificate and reproducibility checks; the external-baseline comparison below reports measured timings under a fixed configuration without asserting an advantage of the native pipeline.

13.1 External baseline comparison on identical instances

A fixed-hardware external-baseline experiment compares the native pipeline with five external solvers on identical finite instances. The recorded configuration is: a two-core Intel Xeon virtual machine with 4.1 GiB of memory, Debian GNU/Linux 13 with kernel 5.10.134 and glibc 2.41, and harness Python 3.12.14; minisat (2.2 series, built from the public mirror at commit 16982a5 with gcc 14.2 at optimisation level `-O3` and the standard compiler-compatibility patch), kissat 4.0.4 (commit 8af8e56, gcc 14.2, `-O3`), z3 5.1.0, CP-SAT from OR-Tools 9.15.6755, and HiGHS 1.15.1; a 60-second wall-clock timeout, enforced as a hard kill at sixty seconds for the subprocess SAT solvers and, for the in-process CP-SAT and HiGHS runners and the bounded native runner, at ninety seconds, the thirty-second grace covering interpreter startup and model construction; a 3 GiB address-space cap; three repetitions per solver and instance with the median reported; single-threaded deterministic configurations throughout, with CP-SAT restricted to one worker and a fixed random seed. The artifact’s protocol block machine-records the processor, core count, memory, kernel version, glibc version, and harness Python; the distribution name, Debian GNU/Linux 13, is stated from the operator’s environment record rather than machine-captured, and is consistent with the recorded glibc 2.41 userland. Timings are whole-process wall clock and include solver startup; the CP-SAT and HiGHS timings additionally include interpreter startup and model construction.

Every instance is generated as a finite ECD presentation and encoded once by the artifact’s encoders, so that all solvers receive the same finite relation data: DIMACS CNF for minisat, kissat, and z3; the LP-format MILP encoding for HiGHS; and the minimal forbidden-tuple encoding for CP-SAT. The families are the controlled-width benchmark families (path disequality chains, binary trees, banded chains at width two, and two-row grids) and the unsatisfiable odd-ring family; instance sizes range from 5 to 2048 policy variables. On each instance the native tree-decomposition dynamic program is run with the supplied decomposition and timed, its certificate is independently re-verified and timed, and for satisfiable instances a policy is reconstructed from the certificate and verified. The experiment covers 24 instances and 120 external solver runs.

Table 6 reports median wall-clock times at the largest instance of each family; the artifact report contains the full per-instance data, including minimum and maximum repetitions and peak memory.

Family	$ X $	clauses	DP	verify	minisat	kissat	z3	CP-SAT	HiGHS
path-neq	2048	4094	0.388	0.682	0.006	0.004	0.007	0.611	0.130
binary-tree-neq	1023	2044	0.135	0.211	0.004	0.003	0.005	0.382	0.122
banded-chain-w2	2048	4092	0.546	0.930	0.006	0.004	0.008	0.418	0.284
grid-neq 2×256	512	1532	0.055	0.087	0.004	0.003	0.004	0.370	0.116
odd-ring-neq	1025	2050	0.169	0.304	0.005	0.003	0.005	0.371	0.121

Table 6: Median whole-process wall-clock times in seconds at the largest instance of each family, as recorded in `reports/external_baseline_sweep.json`. DP and verify are the native dynamic program and its independent certificate re-verification; the odd-ring family is unsatisfiable, the others satisfiable.

The cross-check accounting is complete: all 120 external runs agree with the native certificates; each of the 95 external satisfiable models is parsed, converted to a policy on the original instance, and verified natively; each of the 25 external unsatisfiable answers is checked against an independently verified DP-REFUTATION certificate; no run is inconclusive or rejected. Peak resident memory of any external run is at most 100.2 MiB against the 3 GiB cap. At these certificate-study sizes the compiled SAT solvers complete in milliseconds, with startup dominating the measurement, while the reference native implementation is a Python program whose wall-clock cost is dominated by certificate emission and independent verification rather than search. The experiment therefore supports the reproducibility and cross-checking claims of this article; it does not support, and the article does not make, a performance-superiority claim over the external solvers.

13.2 Comparative scaling study on mechanics instances

The fixed external-baseline protocol is applied unchanged to fourteen Boolean instances derived from the mechanics families: both observation variants of the ten-bar truss, the pre-tensioned chain at 256 and 1024 springs, the dense-influence chain at 8 and 16 actuators, and the lattice at one and two columns. Identical encodings are emitted once; every external run is bounded by the 60-second timeout, the 3 GiB address-space cap, and the single-threaded deterministic configurations; timings are medians of three repetitions. The native pipeline runs as a bounded isolated process that loads the instance, runs the dynamic program, re-verifies its certificate, and reconstructs and verifies a policy when satisfiable, so native and external wall-clock figures cover comparable whole-process pipelines. Certified relation generation and its independent verification are accounted separately from the recorded mechanics benchmark suite and excluded from both sides.

Instance	$ X $	clauses	native	mini.	kissat	z3	CP-SAT	HiGHS
ten-bar, same	4	82	0.07	0.003	0.002	0.003	0.363	0.106
ten-bar, split	8	82	0.12	0.003	0.002	0.012	0.384	0.127
chain $n=256$, same	256	512	0.12	0.005	0.002	0.004	0.388	0.122
chain $n=1024$, same	1024	2048	0.37	0.003	0.002	0.004	0.366	0.115
chain $n=1024$, split	2048	2048	1.02	0.003	0.002	0.005	0.370	0.116
dense $n=8$, same	8	487	0.12	0.004	0.002	0.003	0.358	0.112
dense $n=16$, same	16	130942	7.18	TMO	3.775	1.648	49.5	TMO
dense $n=16$, split	32	130942	14.54	TMO	0.367	0.355	4.289	5.203
lattice $C=1$, same	6	196	0.12	0.003	0.002	0.003	0.372	0.115
lattice $C=2$, same	12	26568	0.92	2.002	0.139	0.474	1.716	6.034
lattice $C=2$, split	24	26568	1.47	1.294	0.034	0.569	0.860	0.946

Table 7: Median whole-process wall-clock times in seconds for the comparative scaling study on mechanics-derived instances (selected cases; the artifact report contains all fourteen). TMO marks runs that exceeded the fixed 60-second limit and are recorded as inconclusive. Native timings cover the whole bounded pipeline including certificate emission and independent re-verification. The recorded study was replicated in full under version-exact solver provenance with an identical summary; see Subsection 13.2.1.

The cross-check accounting over the runs of Table 7 is complete: all fourteen native certificates verify; 34 external satisfiable models are verified natively as policies; 33 external unsatisfiable answers are cross-checked against verified DP-REFUTATION certificates; three external runs exceed the limit and are recorded as inconclusive; nothing is rejected; and all 67 status-bearing external answers agree with the native certificates.¹ Finite-element (FE) preprocessing for the dense in-

¹A forensic audit of the recorded artifact confirms the recorded counters against the per-instance records: the CP-

stances amounts to about 2.7 seconds of certified relation generation and 2.7 seconds of verification, reported separately.

The recorded observations are per-instance and carry no superiority claim. On the width-bounded families every solver answers in milliseconds and the native Python pipeline is slower in wall-clock, consistent with the earlier external-baseline experiment. On the dense-influence family at sixteen actuators, whose forbidden-tuple encoding has 130,942 clauses, minisat exceeds the limit on both observation variants and HiGHS exceeds it on the unsatisfiable variant, while kissat answers in 3.8 seconds, z3 in 1.7 seconds, CP-SAT certifies the unsatisfiable variant in about 50 seconds, and the native dynamic program certifies the refutation in 7.2 seconds of wall-clock including certificate emission and verification. The native certificates on this family grow as 2^n (57 and 116 MB at $n=16$, and 120, 243, and 253 MB at $n=17$ and $n=18$ in the extension below), which is the recorded cost of the certificate-emitting pipeline and the honest boundary of the dense family in this artifact.

13.2.1 Version-exact replication and dense-family extension

Two follow-up studies of the artifact release consolidate the recorded comparative scaling study: a full replication under version-exact solver provenance, and an extension of the dense-influence family beyond the recorded boundary $n=16$.

Full replication. The complete fourteen-instance sweep was re-executed with all five solvers rebuilt or pinned to the recorded versions of Subsection 13.1. The replication reproduces the recorded summary exactly — fourteen verified native certificates, 34 verified satisfiable cross-checks, 33 verified unsatisfiable cross-checks, three inconclusive runs, and 67 agreements — and a field-by-field comparison (`diff_reports.py`, timing and size fields excluded as hardware-dependent) finds 1135 matching leaf fields whose only mismatches are the certificate file names, the per-case provenance notes carried only by the recorded report, and the hardware block’s memory figure (3.9 against 4.1 GiB); the replication’s native runner re-emits byte-identical duplicate certificates, with SHA-256 equality verified for the dense-8 and dense-16 same, ten-bar same, and lattice $C=2$ split certificates and for the three completed dense-17/18 certificates of the extension below. The replicated report is archived in `reports/` as `mechanics_external_baseline_sweep_v2.json`.

Dense-family extension. The dense-influence family is extended to $n=17$ and $n=18$, both observation variants, with the recorded phase-4 driver (the mechanics-benchmark generator) and the fixed protocol unchanged. Generation certifies all four FE row batches and tree decompositions (widths 16, 16, 17, and 17); the dynamic program certifies a refutation at $n=17$ same (120.2 MB certificate), an optimum with a verified policy at $n=17$ split (243.2 MB), and a refutation at $n=18$ same (252.5 MB); at $n=18$ split the certificate-emitting dynamic program exceeds the machine’s physical memory during message construction — the generating process was killed by the kernel out-of-memory (OOM) killer with an anonymous resident set of 3 750 608 kB, about 3.6 GiB, the kernel line being retained with the failure record — and the case is recorded as `DP-MEMORY-LIMIT`

SAT run on the unsatisfiable dense $n=16$ instance answered `UNSAT` in 49.5s with a verified cross-check, not `UNKNOWN`; a version-exact replication reproduces the same answer and the same summary; and the runner code maps a CP-SAT `UNKNOWN` status to inconclusive by construction, so it cannot be bucketed with unsatisfiable answers. The harness agreement counter is gated on verified cross-check evidence, which makes the accounting identity structural. An earlier version of this article reported a “corrected” accounting of 32 cross-checked unsatisfiable answers, 4 inconclusive runs, and 66 agreements, on the mistaken belief that the CP-SAT run had returned `UNKNOWN`; that correction is itself withdrawn.

with its model, certified row batch, instance, and validated decomposition archived. The comparative sweep then applies the fixed external-baseline protocol unchanged to the four extension instances.

Instance	clauses	native	mini.	kissat	z3	CP-SAT	HiGHS
dense $n=17$, same	262002	15.2	TMO	15.95	4.90	UNK	TMO
dense $n=17$, split	262002	30.9	TMO	0.81	0.75	8.81	13.02
dense $n=18$, same	524123	31.6	TMO	52.18	TMO	TMO	TMO
dense $n=18$, split	524123	MEM	TMO	3.30	1.71	UNK	29.89

Table 8: Median whole-process wall-clock times in seconds for the dense-family extension at $n=17$ and $n=18$. Native timings cover the whole bounded pipeline including certificate emission and independent re-verification; TMO marks runs that ended at or beyond the wall-clock limits without a recognised status; UNK marks CP-SAT runs that returned UNKNOWN; MEM marks the bounded native run that raised **MemoryError** at the 3 GiB address-space cap; all marked runs are recorded as inconclusive. The limits are two-tier as in the recorded study: the in-process CP-SAT and HiGHS runners carry a sixty-second solver limit with a ninety-second hard process cap whose grace covers interpreter startup and model construction. Every status-bearing external answer is cross-checked: unsatisfiable answers against the verified DP-REFUTATION certificates of the same cases, satisfiable models as natively verified policies.

As in the recorded study, the observations are per-instance and carry no superiority claim. minisat exceeds the limit on all four extension instances; CP-SAT ends without a recognised status on the three unsatisfiable or encoding-heaviest variants, whose forbidden-tuple encodings comprise 137.7–388.1 MB of JSON: it returns UNKNOWN within its solver limit on the $n=17$ same variant (40 s), overshoots the soft sixty-second solver limit before returning UNKNOWN on the $n=18$ split variant (73 s of solve time), and is killed at the ninety-second hard process cap on the $n=18$ same variant; kissat and z3 answer both $n=17$ variants in seconds; kissat is the only solver answering the unsatisfiable $n=18$ variant within the limit (52.2 s); and on the satisfiable $n=18$ split variant kissat, z3, and HiGHS return models that verify natively as policies while the certificate-emitting native pipeline cannot complete within the same 3 GiB address-space cap. The cross-check accounting over the twenty extension runs of Table 8 records seven verified satisfiable models, three verified unsatisfiable answers, ten inconclusive runs, no rejections, and seven agreements with the native certificates — the three verified satisfiable answers at $n=18$ split have no native status to agree with, since the native run is inconclusive at the memory cap. FE preprocessing is accounted separately as in the recorded study, and the extension reports (`mechanics_dense_extension.json`, `mechanics_dense_extension_sweep.json`) are archived with the artifact.

The artifact, including the prototype, the reproduction scripts, the recorded reports, the replication and dense-extension reports of Subsection 13.2.1, and the release manifest with the full solver build provenance, is archived in the project repository <https://github.com/MIKEAA2020/halting-physics> under release tag `halting_physics_workspace_v4`. Extension artifacts above the repository’s per-file size limit are archived by SHA-256 checksum and size in the release manifest and are exactly regenerable from the archived instances and decompositions; the small artifacts are deposited with their checksums; the archive mechanics are recorded in Appendix A. An archival DOI will be assigned before publication.

14 Limitations and conclusions

The companion theory is deliberately finite and certificate-driven. It does not claim a universal dichotomy for all finite ECD instances, practical superiority over generic solvers, or continuum-mechanics validity from finite algebraic row certificates alone. More specifically, it does not claim that inner-model infeasibility proves physical infeasibility, that solver timeout proves bounded non-realizability, that the sparsification theorem is practically useful on every mechanical model, or that the D^P [20], ETH [14], $W[1]$ and $W[2]$ [10] frontiers have been proved for every finite ECD layer; the $W[2]$ -hardness of sensor repair with its conditional approximation and ETH corollaries, and the classification of unrestricted sensor synthesis (Propositions 7.3 and 7.4), are the design-side results this article records — problems rather than a layer. Those statements require additional reductions, modelling hypotheses, or experiments. The recorded mechanics study adds its own boundaries: the structural front end certifies declared rational models, not discretisation error; the dense-influence families exhaust the relation-table and certificate-size limits of the pipeline at moderate sizes; the sparsification ladders certify monotonicity while remaining inconclusive at reduced radii on the calibrated families; and the comparative scaling study records per-instance observations only, in which external solvers and the native pipeline each fail and succeed on different instances of one family under the fixed protocol.

The positive claim is narrower and checkable: once a finite source representation is declared, the solver and verifier can produce typed evidence for policies, refutations, optimum costs, sensor repairs, finite mechanical relations, and sparsification ladders. The location of failure remains part of the presentation, and the certificate records the data needed to transport that claim.

A Artifact, certificate formats, and reproducibility

This appendix describes the finite ECD prototype accompanying the finite algorithmic and mechanics development. The artifact is a correctness and certificate harness for finite explicit-table instances, exact finite affine algebraic examples, and declared exact rational structural models. It is not a continuum-mechanics validator, and its external-baseline comparisons are cross-check studies rather than performance-superiority claims.

A.1 Finite instance format

A finite explicit-table ECD instance is encoded as a JSON object with the fields `variables`, `domains`, and `factors`. A typical Boolean example is

```
{
  "variables": ["x0", "x1"],
  "domains": {"x0": [0, 1], "x1": [0, 1]},
  "value_costs": {"x0": {"0": 0, "1": 1}},
  "factors": [
    {"name": "neq", "scope": ["x0", "x1"],
     "relation": [[0, 1], [1, 0]]}
  ]
}
```

The field `variables` fixes the ordered policy variables; `domains` gives each finite domain; `factors` gives explicit permitted tuples on ordered scopes; and `value_costs`, when present, supplies the non-negative value costs κ of the article’s value-cost remark (Remark 2.6) for optimum certificates. The

complexity and certificate claims made for this artifact are explicit-table claims unless a different generator is specified.

A.2 Native certificate schemas

The verifier recognises several certificate types. A **POLICY** certificate assigns one domain value to each variable and is checked by scanning all factor relations. A **FIBRE-CONFLICT** certificate names an observation fibre and, for each action, a scenario in that fibre excluding the action. A **SENSOR-REPAIR** certificate records a selected sensor set for the small one-shot repair examples; the optimality certificates are exhaustive certificates for small instances.

A **TREE-DECOMP** certificate contains bags, tree edges, a root bag, and a factor-to-bag map:

```
{
  "type": "TREE-DECOMP",
  "root": "b0",
  "bags": {"b0": ["x0", "x1"]},
  "edges": [],
  "factor_to_bag": {"neq": "b0"}
}
```

The verifier checks factor containment, tree connectivity, and the running-intersection property. A **DP-OPTIMUM** or **DP-REFUTATION** certificate contains a tree decomposition, dynamic-programming message tables, and a root optimum or refutation. The verifier recomputes all messages; solver logs are not treated as certificates. Gate outputs use the bookkeeping markers of the article’s status discipline (Remark 3.1) — **VERIFIED**, **REJECTED**, **COMPLETE**, and **RELATION-TABLE-LIMIT** — which record the state of a harness check and are distinct from the four mathematical answer statuses.

For exact affine mechanics-style examples, an **FE-ROW** certificate records the row name, action assignment, exact response interval, safety interval, and classification as **SAFE** or **UNSAFE**; exact rational rows are always one of these two. Interval-valued rows, arising from residual-bounded rather than exact arithmetic, additionally admit **UNRESOLVED**, which occurs exactly when the certified response interval crosses the declared safety threshold. An **FE-ROW-BATCH** collects such rows and includes the derived finite ECD relation-table instance. A **SPARSIFICATION-BATCH** records retained scopes, exact omitted-term intervals, retained-response intervals, and the corresponding inner and outer sparse finite instances. A sparsification-ladder check compares two such batches for retained scopes $J \subseteq J'$: coarse inner cylinders (a cylinder is the set of completions of one retained assignment) must refine into fine inner rows, and fine outer rows must project into coarse outer rows.

A.3 Declared structural models and mechanics certificates

The mechanics layer adds declared structural models in the sense of the article’s mechanics front end (Subsection 8.1), in two kinds. A truss model lists nodes with rational coordinates and support flags, members with declared rational axial stiffnesses, actuators (node-force patterns or self-equilibrated axial member-pair patterns), load scenarios, and monitors (member forces or displacement components). A scalar-chain model lists grounded spring stiffnesses with the same actuator and monitor types.

A **MECH-SOLVE** certificate records a scenario’s load vector and exact displacement vector over the free degrees of freedom. The verifier re-assembles the stiffness matrix with independently written assembly code and checks $K\hat{u} = f$ exactly by substitution. A **MECH-ROW-BATCH** records, for every scenario and monitor, the exact rational constant term and sparse actuator influence coefficients,

together with the derived relation table over the row’s support; the verifier regenerates every row and relation through its own assembly and solve code and compares. Rows whose support exceeds the recorded relation-enumeration cap are marked `RELATION-TABLE-LIMIT`. A `MECH-SPARSIFY-BATCH` records retained scopes, exact omitted intervals, inner and outer row counts, and the inner and outer finite instances; the ladder check verifies nesting, the inner-cylinder inclusion (through the two per-term extreme completions, which are binding because the response is linear in the omitted actuator values), and the outer projection.

Instances derived from structural models come in the two observation variants (`same` and `split`) of the article’s benchmark convention (Section 11). Decompositions follow the deterministic minimum-fill heuristic and tree-decomposition validation of the same suite.

A.4 Baseline encodings and external solvers

The script `baseline_encode.py` emits DIMACS CNF, LP-format MILP, and a minimal CP-SAT-style forbidden-tuple JSON encoding for Boolean explicit-table instances. These encodings fix identical finite instances for external comparisons: the DIMACS encoding is consumed by the SAT solvers and `z3`, the LP-format encoding by the MILP solver, and the JSON encoding by the CP-SAT runner.

External solver output is handled conservatively. The optional solver adapters record availability, timeouts, and raw status. The cross-checker accepts an external SAT or UNSAT result only when an independent native ECD certificate is also verified: SAT requires a policy certificate, and UNSAT requires a verified `DP-REFUTATION`. Missing solvers, timeouts, solver statuses without a certified claim such as a CP-SAT `UNKNOWN` returned within the limit, and unrecognised output are reported as `INCONCLUSIVE`.

The external-baseline experiments of the article are executed by the sweep scripts and isolated runners

```
external_baseline_sweep.py, mechanics_baseline_sweep.py,
cpsat_runner.py, highs_runner.py, native_dp_runner.py.
```

The protocol is the fixed external-baseline protocol of the article (Subsection 13.1). In the mechanics study the native dynamic program runs as a bounded isolated process that emits and re-verifies its certificate, and FE preprocessing is accounted separately from the phase 4 report — the recorded report of the mechanics-benchmark generation phase — rather than charged to either side. Solvers are located on the path or through the environment variables

```
EXTERNAL_ECD_MINISAT, EXTERNAL_ECD_KISSAT,
EXTERNAL_ECD_Z3, EXTERNAL_ECD_CPSAT_PYTHON,
EXTERNAL_ECD_HIGHS_PYTHON;
```

with no solver available the emitted report is `INCONCLUSIVE`. External answers are cross-checked under the conservative policy above. The build provenance of every solver (source commit, compiler, and patches) is recorded in the release manifest. The summary counters are audited against the per-instance records by the deposited audit tool, and the `status_agreement_with_native` counter is gated on verified cross-check evidence: an external run counts as an agreement with the native pipeline only when its classification matches the native status and its cross-check status is `VERIFIED-SAT` or `VERIFIED-UNSAT`, so a rejected or evidence-free answer can never inflate the agreement count.

The recorded comparative scaling study was replicated in full under the version-exact solver provenance of the article’s baseline configuration (Subsection 13.1); the replicated report is archived as `reports/mechanics_external_baseline_sweep_v2.json` and reproduces the recorded summary and every structural field, with byte-identical re-emitted native certificates.

A.5 Reproduction commands

From the artifact directory, the repository-ready entry point is

```
make reproduce
```

or equivalently

```
./reproduce.sh
```

This runs the smoke checks, the controlled-width benchmark sweep, the SVG plot generation, the gated external-baseline sweep, the mechanics benchmark suite, the width-adaptive sparsification study, and the comparative scaling study on mechanics-derived instances. The principal outputs are

```
repro_report.json, reports/benchmark_sweep.json,  
reports/certificate_size_vs_width.svg,  
reports/verify_time_vs_certificate_size.svg,  
reports/external_baseline_sweep.json,  
reports/mechanics_benchmarks.json,  
reports/sparsification_study.json, and  
reports/mechanics_external_baseline_sweep.json.
```

The replication report `reports/mechanics_external_baseline_sweep_v2.json` and the dense-extension reports of Subsection A.6 are recorded alongside these outputs. The recorded mechanics baseline sweep was executed in case chunks with the `--only` and `--merge` options of `mechanics_baseline_sweep.py`, which is equivalent to a single full invocation; the chunked invocation is documented in the release manifest. The dense-chain $n = 16$ baseline encodings and the byte-identical duplicate native certificates are regenerable and are not archived; the study report references the archived phase 4 certificates.

A.6 Dense-family extension and artifact archive policy

The dense-influence family is extended to $n = 17$ and $n = 18$ (both observation variants) by two checkpointed drivers deposited with the artifact:

```
python3 dense_extension_generate.py --case dense-chain|17|same  
python3 dense_extension_generate.py --assemble  
python3 dense_extension_sweep.py --cases <family|size|obs>  
                                --solvers <names>  
python3 dense_extension_sweep.py --finalize
```

Generation uses the recorded phase-4 driver unchanged (full relation enumeration, deterministic minimum-fill decomposition, certified DP with policy reconstruction); the sweep applies the fixed external-baseline protocol unchanged. Both drivers checkpoint their progress after every case or (case, solver) unit because the recording environment terminates long-running sessions, so each unit is an unmodified invocation of the recorded runner functions under the fixed limits. The principal outputs are

```
reports/mechanics_dense_extension.json,  
reports/mechanics_dense_extension_sweep.json,  
reports/dense_ext_gen_*.json, and  
reports/dense_ext_sweep_progress.json;
```

the latter two hold the per-case generation records (including the `_FAILED` record of the $n = 18$ split memory-limit case) and the per-unit sweep checkpoints.

The extension produces the honest boundary records of the study: the DP certificate sizes 120.2, 243.2, and 252.5 MB at $n = 17/18$, and the $n = 18$ split case whose certificate-emitting DP exceeds the machine’s physical memory (recorded as `DP-MEMORY-LIMIT`, with the model, certified row batch, instance, and validated decomposition archived). The bounded sweep run of the same case raises `MemoryError` at the 3 GiB address-space cap and is recorded as inconclusive.

Extension artifacts above the repository’s 100 MB per-file limit — the Dense DIMACS, LP, and CP-SAT encodings and the DP certificates — are archived by SHA-256 checksum and size in `reports/dense_extension_large_artifacts.json` (built by `build_dense_ext_manifest.py`) and are exactly regenerable from the archived instances and decompositions; the native runner re-emits the DP certificates byte-identically. The small artifacts of every extension case — instance, model, certified row batch, validated decomposition, native run records, the $n = 17$ split policy certificate, and the minisat result files — are deposited in `examples/` with their checksums appended to `examples/SHA256SUMS`.

A.7 Claim boundary

The artifact certifies finite explicit-table examples, exact finite affine algebraic models, and declared exact rational structural models (pin-jointed trusses with rational directions and grounded mass-spring chains). It does not certify continuum discretisation error, physical modelling adequacy, or practical superiority over external solvers. External baseline results are reported only under a fixed configuration of solver versions, hardware, time limits, memory limits, and encodings, as recorded in the release manifest; timeouts remain `INCONCLUSIVE`. The sparsification study certifies ladder soundness and monotonicity and records the measured width–inconclusiveness tradeoff; it does not claim a practical width-reduction benefit on the calibrated families. The dense-family extension records the certificate-growth cost (2^n certificates of up to 252.5 MB) and the memory boundary of the certificate-emitting pipeline at $n = 18$; both are properties of this artifact’s environment and protocol, not performance claims about the underlying algorithms.

B Standard reductions for the explicit-table frontier

This appendix records the encodings and standard reductions behind the complexity table of Section 5, in the explicit-table input format of Section 2. The membership parts are immediate from the format; the reductions are standard and are stated at the encoding level so that the verifier-facing representation is unchanged. All reductions satisfy the nonempty-local-relation promise: no factor table of the constructed instance is empty.

B.1 Tractable rows

Verifying a proposed assignment scans the factor tables, so it is linear in the input size. Unary constraints filter the domains variable by variable. Every Boolean binary factor is a conjunction of 2-clauses, one per forbidden tuple: forbidding (v_1, v_2) on (x, y) is the clause $(x \neq v_1) \vee (y \neq v_2)$. Boolean binary instances are therefore 2-SAT formulas and are decided in linear time [1].

B.2 Hard rows

Boolean ternary factors are NP-complete: a 3-CNF clause $(\ell_1 \vee \ell_2 \vee \ell_3)$ becomes a factor on its three variables whose table permits the seven satisfying tuples, and the satisfying assignments of the formula are exactly the global assignments of the instance. Membership in NP is witnessed by the assignment. The fixed-template background is classified by the dichotomy theorems [23, 27].

Binary disequality over a three-element domain is NP-complete by graph 3-colourability: vertices become variables with domain $\{1, 2, 3\}$, and edges become disequality factors whose tables permit six of nine tuples [12].

Nonexistence of a global assignment is coNP-complete: it is the complement of the ternary satisfiability problem above, and a satisfying assignment is the polynomial witness for existence.

Counting global assignments is #P-complete: the same clause-to-factor encoding reduces #3-SAT, and counting the satisfying assignments of the formula is counting the global assignments of the instance [24, 6].

Minimum-cardinality fibre obstruction is NP-hard from set cover [16]. Given a universe A and sets $X_1, \dots, X_n \subseteq A$, build one observation fibre with action set A and one scenario w_j per set X_j with admissible set $S_{w_j} = A \setminus X_j$. A set C of scenarios has empty admissible intersection if and only if $\bigcup_{w_j \in C} X_j = A$, so minimum-cardinality conflict cores are exactly minimum set covers. The local relations are nonempty whenever $X_j \neq A$, which holds after discarding redundant set rows.

Minimum-cardinality sensor repair from a listed candidate family is NP-complete from set cover [16]. Given a universe U , covering sets $X_1, \dots, X_n \subseteq U$, and a budget k , let $W = \{w_0\} \cup \{w_j : j \in U\}$ carry a single observation value, let $A = \{0, 1\}$ with $S_{w_0} = \{0\}$ and $S_{w_j} = \{1\}$, and let the candidate family contain, for each covering set X_i , the sensor s_i with $s_i(w_0) = 0$, $s_i(w_j) = 1$ for $j \in X_i$, and $s_i(w_j) = 0$ otherwise. Every conflict contains w_0 and at least one w_j : any nonempty set of scenarios within $W \setminus \{w_0\}$ intersects to $\{1\}$, and $\{w_0\}$ alone intersects to $\{0\}$. The inclusion-minimal conflict cores are therefore exactly the pairs $\{w_0, w_j\}$. The sensor s_i is nonconstant on $\{w_0, w_j\}$ exactly for $j \in X_i$, so by Theorem 7.1 a subfamily of at most k candidates repairs exactly when the corresponding covering sets cover U . The admissible sets are singletons, so the nonempty-local-relation promise holds. The construction preserves the budget, and set cover is W[2]-complete under the standard parameterisation by the number of chosen sets [10], so sensor repair parameterised by the number of added sensors is W[2]-hard. Membership in NP is witnessed by the chosen subfamily together with a selector table for the refined problem, checkable by a scan over the scenario list. The candidate list is part of the input; unrestricted sensor synthesis, in which sensors are not drawn from a declared list, is classified by Proposition 7.4 and Subsection B.3.

B.3 Unrestricted sensor synthesis

Proposition 7.4 reduces unrestricted synthesis under a fixed value alphabet to per-fibre transversal bounds. For the hardness claim in part (3), reduce from hitting set. Given a family $F_1, \dots, F_n$ of nonempty subsets of a universe A_0 and a bound $k \geq 2$, let $t = \lceil \log_2 k \rceil$ and $K_0 = 2^t - k$, so $0 \leq K_0 < 2^{t-1}$. Build one-shot data with a single observation value, action set $A = A_0 \cup \{p_1, \dots, p_{K_0}\}$ with fresh actions p_j , one scenario w_i per set with $S_{w_i} = F_i$, and K_0 further scenarios u_j with $S_{u_j} = \{p_j\}$. Every transversal of the constructed family contains every p_j , and its A_0 -part must meet every F_i , so the transversal number of the constructed family is exactly $\tau(F_1, \dots, F_n) + K_0$. By Proposition 7.4(2), t sensors with values in a two-element set repair if and only if this number is at most $2^t = k + K_0$, that is, if and only if the original family admits a transversal of cardinality at most k . Hitting set is NP-complete and W[2]-complete under the parameterisation by the number of chosen elements, in each case by the standard duality with set cover [16, 10]; the construction is

a parameterised reduction with sensor count $t \leq \log_2 k + 1$, so the decision problem is NP-complete and W[2]-hard in the sensor count. The single-channel presentation of the constructed instance again has one unary factor per scenario and an edgeless primal graph, so the hardness is present at width zero on the unrestricted side as well. For fixed m the characterisation is algorithmic: enumerating the at most $|A|^{a^m}$ candidate transversals per fibre decides whether m sensors exist, without enumerating the exponentially many sensor tables.

B.4 Conditional statements for listed repair

Two standard statements transfer to listed repair through the budget-preserving bijection of the set-cover construction in Subsection B.2, under which repairing subfamilies of at most k candidates correspond exactly to covers of at most k covering sets. First, minimum-cardinality listed repair, on instances whose inclusion-minimal core list, of size N , is part of the finite master problem, is approximable within the harmonic bound $H_N \leq 1 + \ln N$ by the greedy set-cover algorithm applied to the cores [15]; and unless $P = NP$ no $(1 - \varepsilon) \ln N$ -approximation exists, since such an algorithm would transfer back through the bijection to set cover on the constructed instances [9]. Second, dominating set admits no $f(k) n^{o(k)}$ algorithm under the exponential-time hypothesis [14, 7]; the closed-neighbourhood encoding maps a dominating-set instance with parameter k to a set-cover instance with the same parameter and polynomial size, and the construction in Subsection B.2 preserves the budget, so an $|I|^{o(k)}$ algorithm for listed repair would compose with this chain to violate the dominating-set barrier. Under the exponential-time hypothesis, therefore, no $|I|^{o(k)}$ algorithm decides listed repair; the trivial enumeration of subfamilies of size at most k runs in $|I|^{O(k)}$, so the barrier is tight up to the constant in the exponent. Both statements are conditional and single-problem; no approximation or ETH frontier is claimed for any other layer of the explicit-table model.

B.5 Source hypotheses for the conditional frontier

The items invoked in Section 5 only with their exact source hypotheses are: the fixed-template dichotomy theorems [23, 27]; the exponential-time hypothesis for SAT lower bounds [14]; and the W[1] and W[2] hierarchies under the standard parameterisations [10]. None of these is restated as a theorem about the variable-template explicit-table model of this article.

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